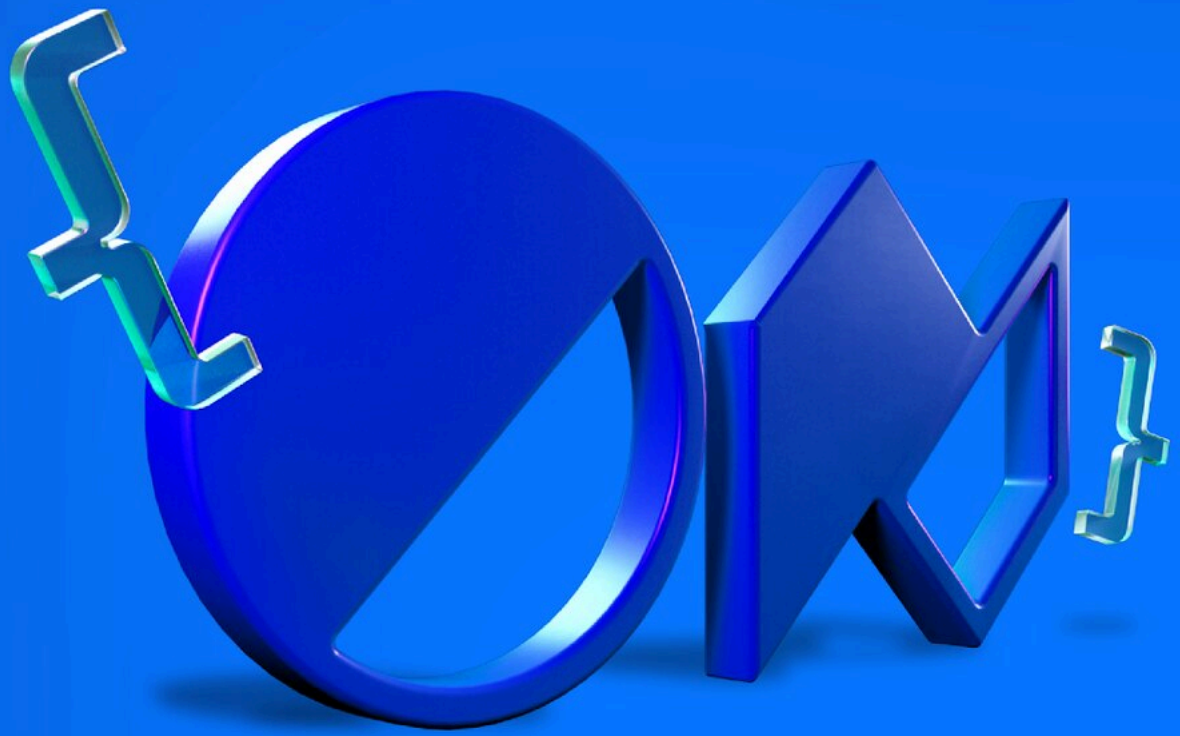
Search is test-time compute

ElasticON AI San Francisco, November 4, 2026

Han Xiao, VP of AI, Elastic



elastic

Search has mostly improved by **training harder**.

Encoders, data, indexes and rerankers all improve before the first query.



Recent LLMs found a second axis: compute at inference

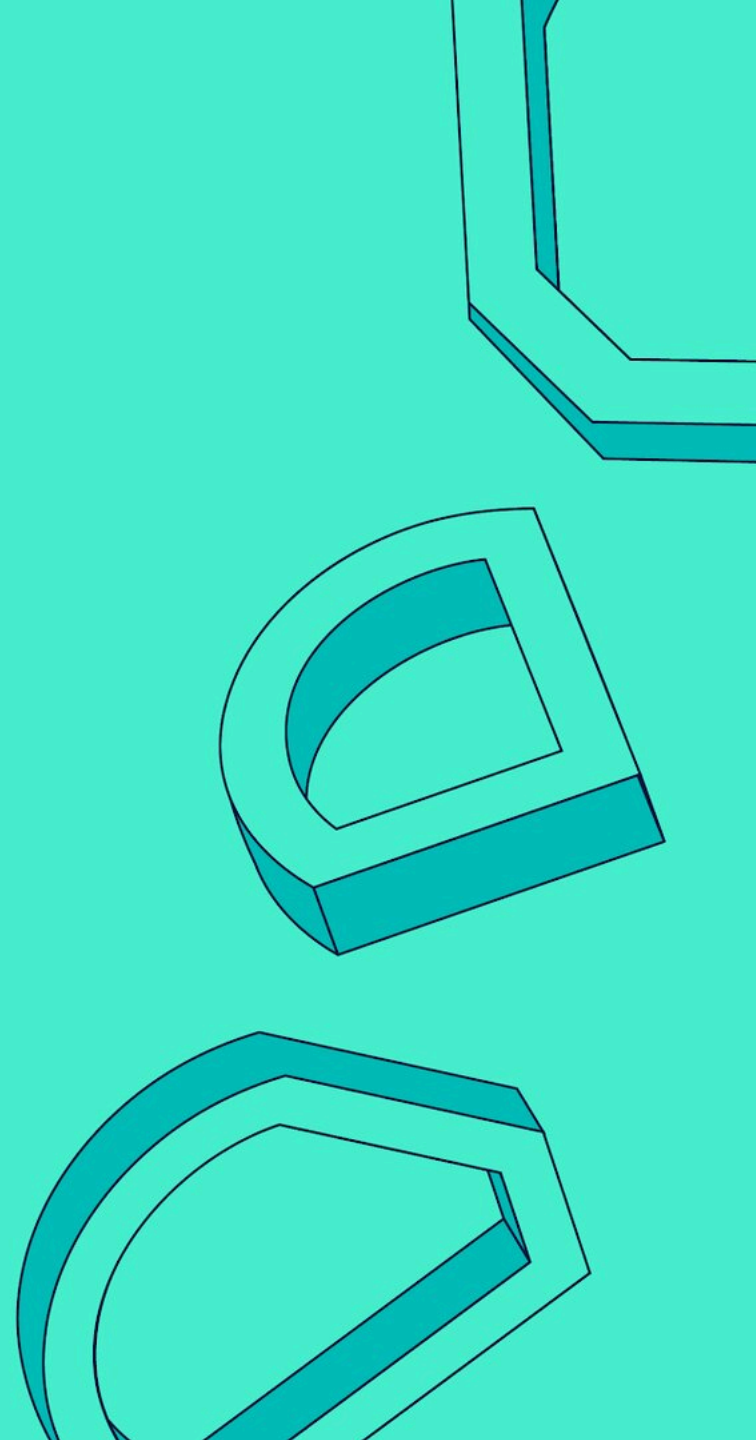
With test-time scaling, the same model does more work per query and quality rises with the budget.



> Having a bot think for just **20 seconds** in a hand of poker got the same performance boost as scaling up the model by **100,000x** and training it for 100,000 times longer.
>
> **Noam Brown, OpenAI, 2024**

Fun fact: search has always lived on **that axis**.

Embedding, retrieval, reranking, expansion and fusion all run after the user presses Enter.



Inference time is a currency, not a KPI

Until now: a KPI

Inference time is a KPI

Latency is a limit written into the SLA. **Better accuracy requires a bigger model.**

becomes

In this talk: a currency

Inference time is a currency

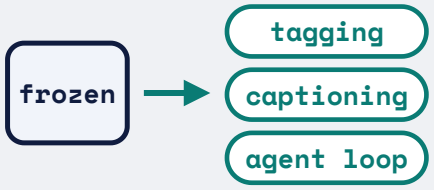
Keep the weights and raise the budget from 10 ms to 100 ms. What does it buy?



buys

Relevance

Better ranking on the same task.



buys

A new capability

Tasks the model was never trained for.

Four training-free demos

Each one buys relevance or a new capability with compute at test time.



01

Retrieval programs

Higher relevance from the same frozen encoder.



02

Open-vocabulary tagger

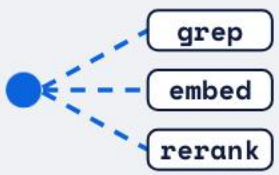
Annotate photos with any words.



03

Semantic highlighting

A reranker highlights the passages that match.



04

Reusable search pipelines

An agent builds small pipelines and reuses them in long-horizon tasks.

elastic

Three ways to spend compute on a frozen model

All three keep the weights fixed and differ in where new information comes from.

A Deeper pass



Read every token or patch vector of the pass you already ran.

New information: from inside the pass

B More passes



Run the encoder again on new views, such as sentences or image crops.

New information: from the input

C Calibration



Transform existing vectors with whitening, label propagation or optimal transport.

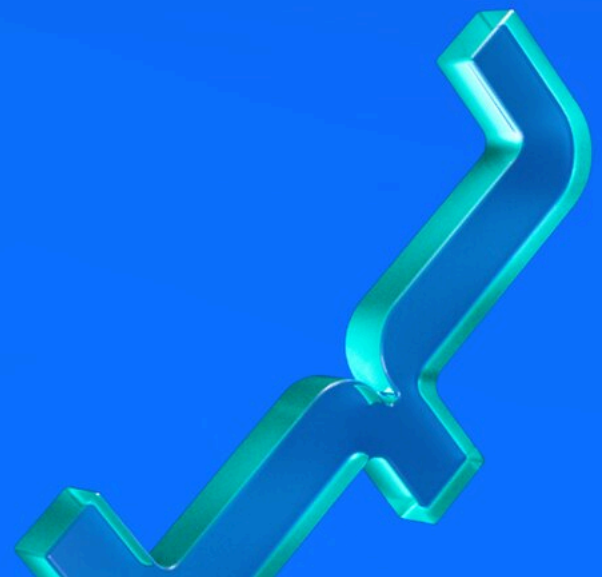
New information: from a global prior

elastic



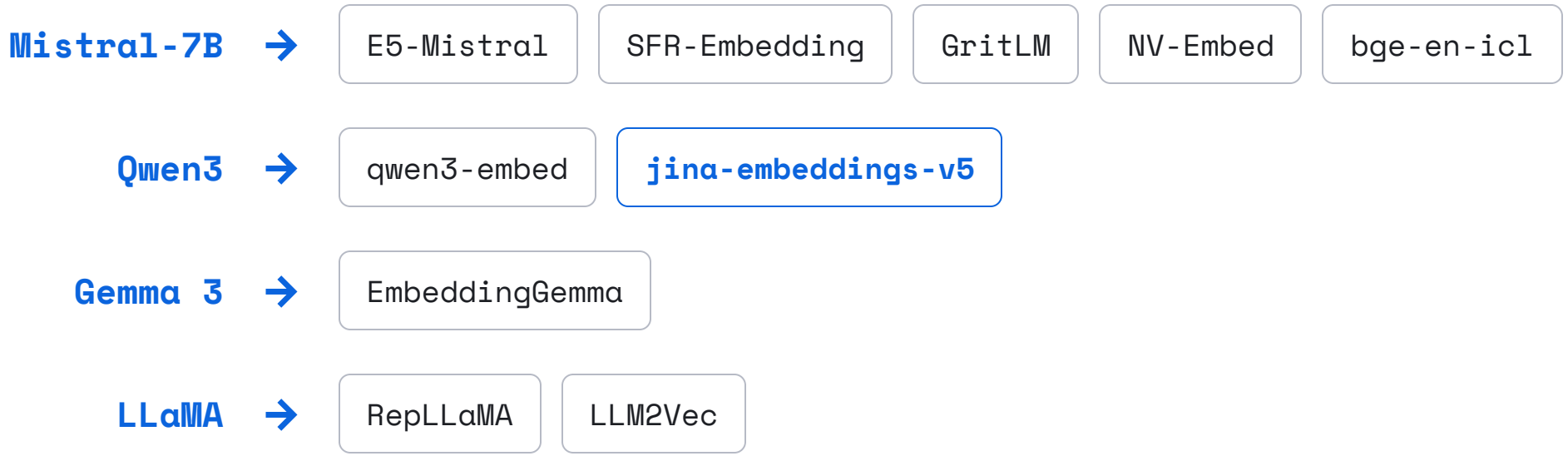
01

Programs on a frozen encoder



Current small embedders are distilled from LLMs

If test-time compute works in the LLM representation space, distilled encoders may inherit it.

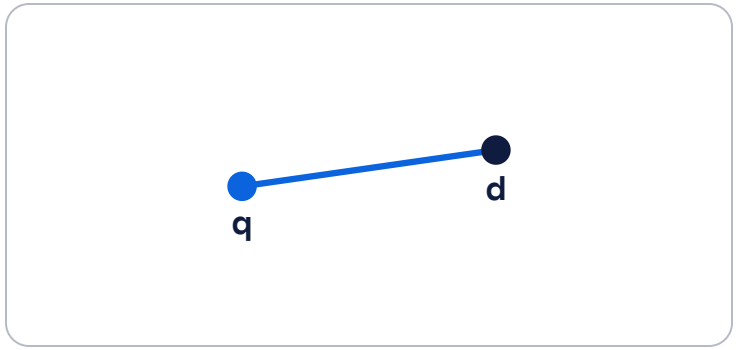


First raised in **jina.ai/news/scaling-test-time-compute-for-embedding-models**

Scoring runs from one cosine to late interaction

A single-vector encoder can split a document into sentences and keep the best match.

single-vector cosine

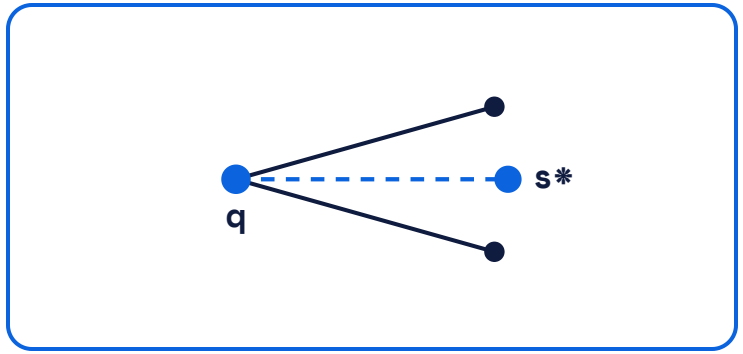


$$\cos(q,\, d)$$

one vector per document

frozen baseline

sentence MaxSim

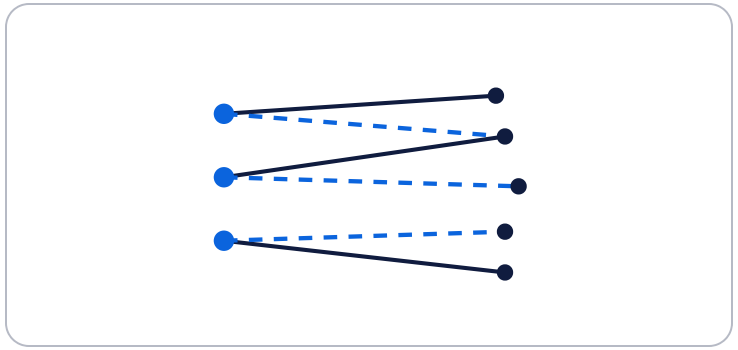


$$\max_s \ \cos(q,\, s)$$

same encoder, one vector per sentence

test-time compute

ColBERT multi-vector



$$\sum_i \max_j \ \cos(q_i,\, d_j)$$

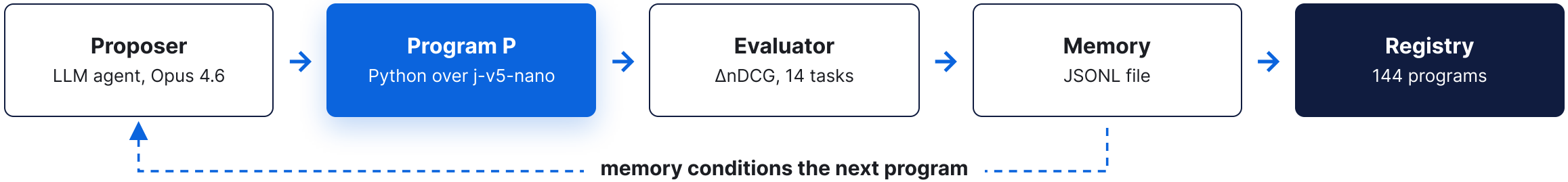
max over document tokens for each query token

needs a multi-vector model

Rules: only the frozen encoder in the query path, with no LLM, second retriever or learned parameters. One **embed_fn** call is one unit of compute.

An LLM agent writes programs over the frozen encoder

Each round changes one program, evaluates it at a fixed budget and keeps it only if the metric improves.



writes	encoder	scores	logs	keeps
one program per round	239M frozen	14 MMTEB tasks ΔnDCG@10	Δ, cost, parent and lesson	144 scored programs

> The evaluator rejects hyperparameters, task-specific routing, external models and learned parameters.

We search on one encoder and test on three others

Discovery · the loop searches here

`j-v5-nano`
239M · Jina
discovery encoder

small and fast, so each round is quick

searched over
14 MMTEB Tier-1 tasks · legal, financial, long-doc, general



`the same program`

Held-out · the loop never sees these

`j-v5-small`
568M · Jina
same family

`gemma-300m`
303M · Gemma 3
unseen family

`qwen3-0.6b`
600M · Qwen3
unseen family

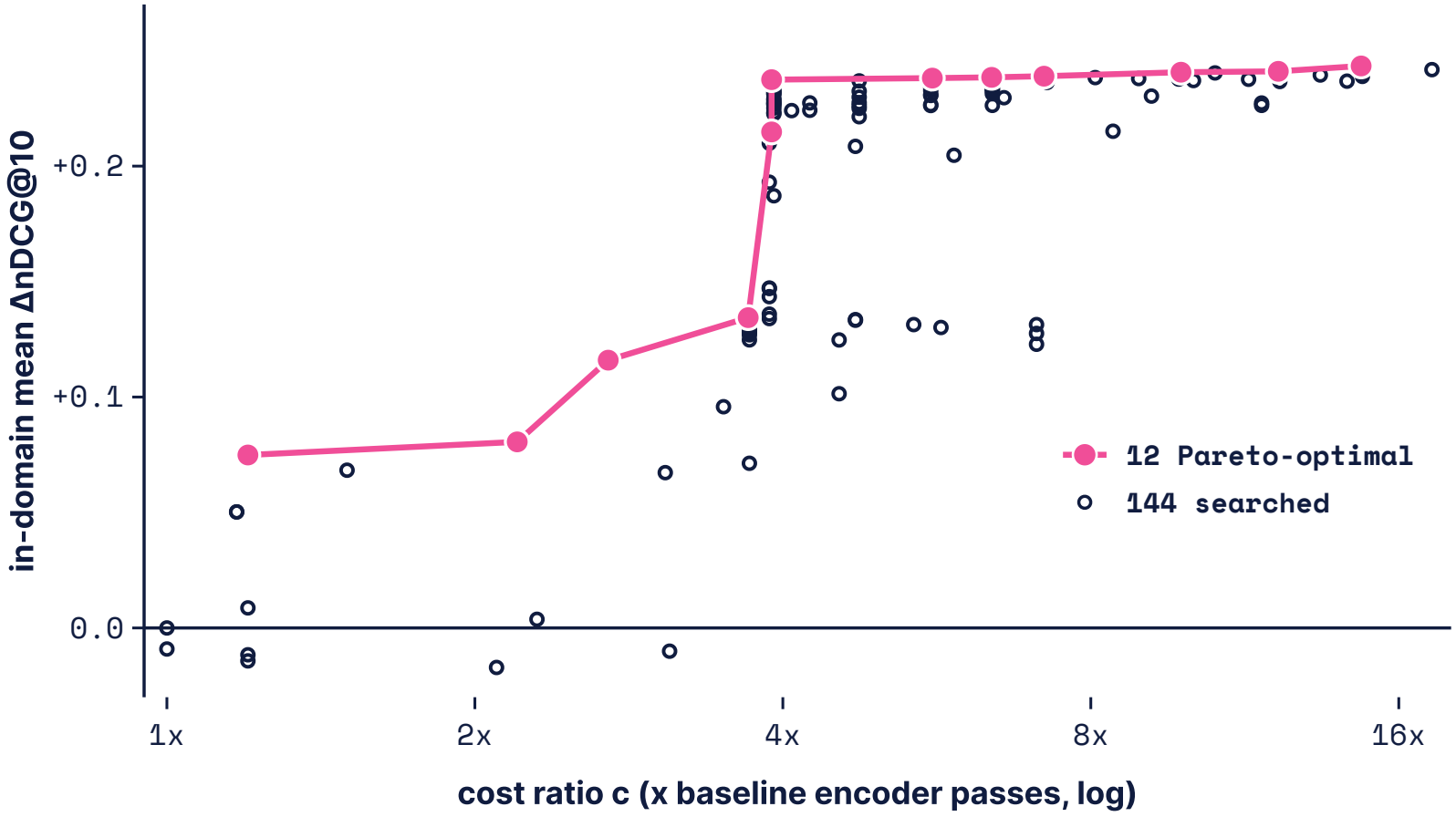
unseen families test transfer

tested on
19 MMTEB Tier-2/3 tasks · summarization, QA, fact-check, ...

Metric: ΔnDCG@10 against the cosine baseline. **Cost c:** encoder passes per query, relative to the baseline.

Rewarding compute gives a Pareto curve

Over 144 programs, the in-domain mean ΔnDCG@10 climbs from +0.07 to +0.24.



144

programs searched

12

Pareto-optimal, from c = 1.2 to 14.7

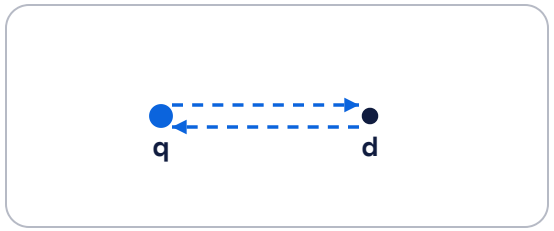
+0.24

best in-domain mean ΔnDCG@10

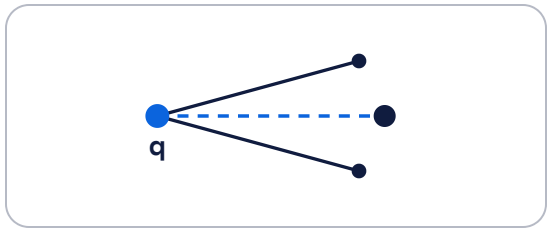
Measured on the 14 discovery tasks.

Twelve programs on the compute frontier

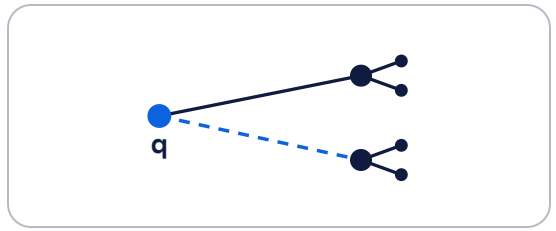
Each recombines the same frozen vectors, at 1.2x to 14.7x the baseline cost.



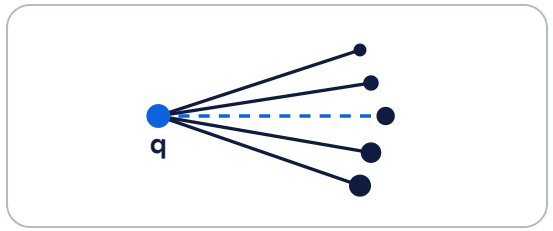
BidirZScore c=1.2



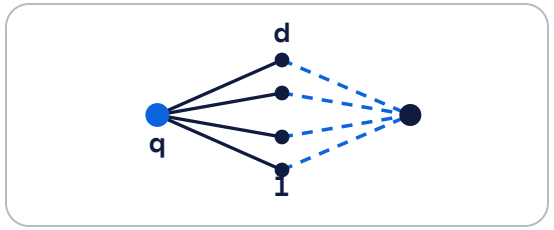
SentMaxSim c=2.2



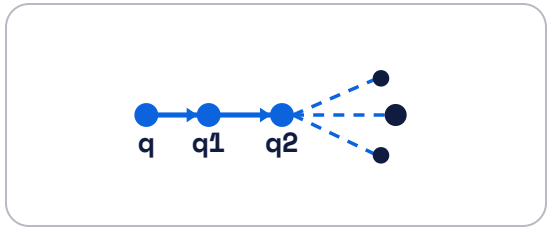
AdaptGranularity c=2.7



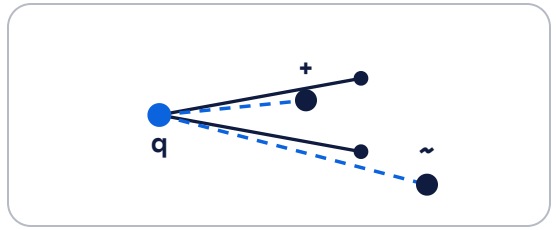
CoverageTriple c=3.7



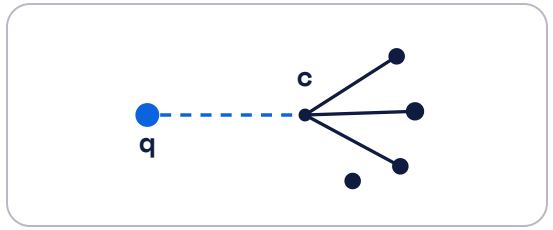
LexicalHybridRRF c=3.9



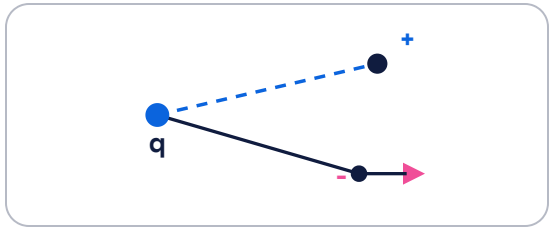
CrossRoundRRF c=3.9



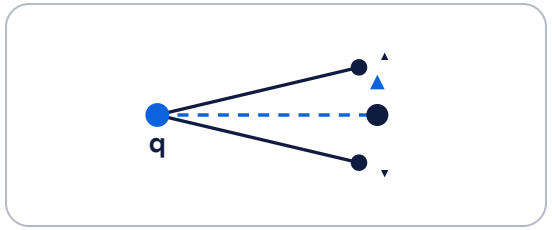
DiverseDualCtx c=5.6



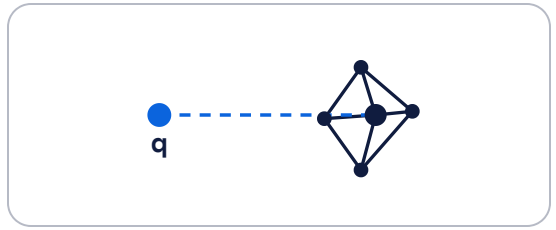
ConsensusRocchio c=6.4



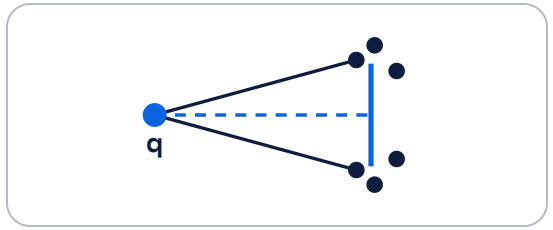
NegContrastive c=7.2



MomentumProg c=9.8

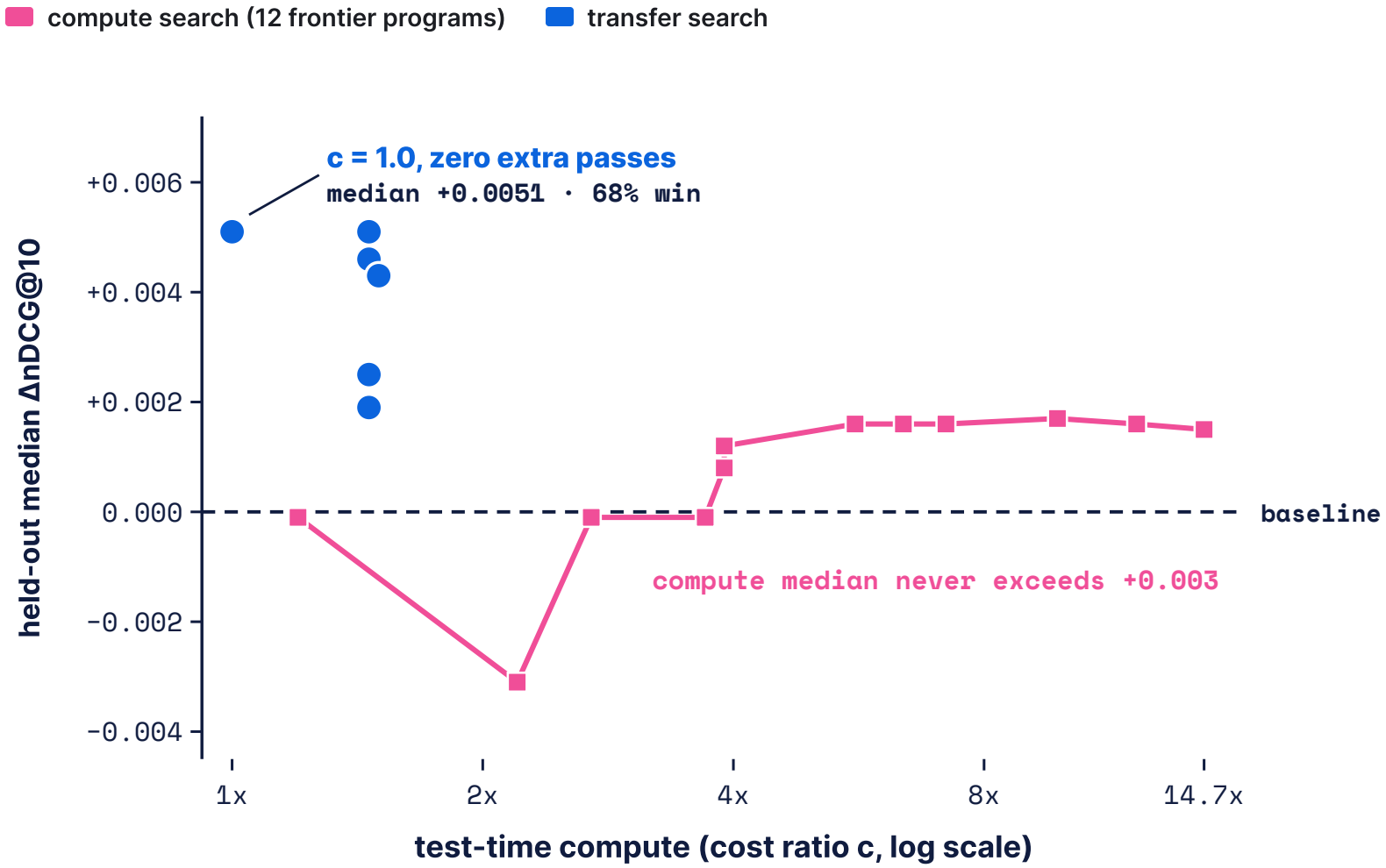


GraphCentrality c=12.2



FisherStability c=14.7

On held-out tasks, a program at c = 1.0 beats all 12 compute programs



+0.003

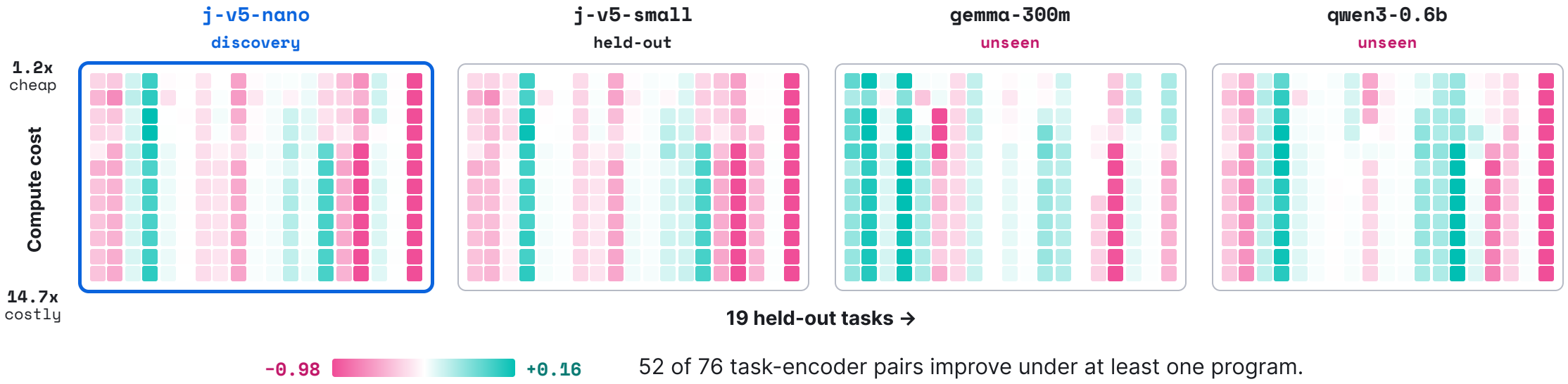
highest held-out median of the compute frontier, up to 14.7x

+0.0051

held-out median at c = 1.0, with a 68% win rate

Compute improves about half the cells and badly hurts others

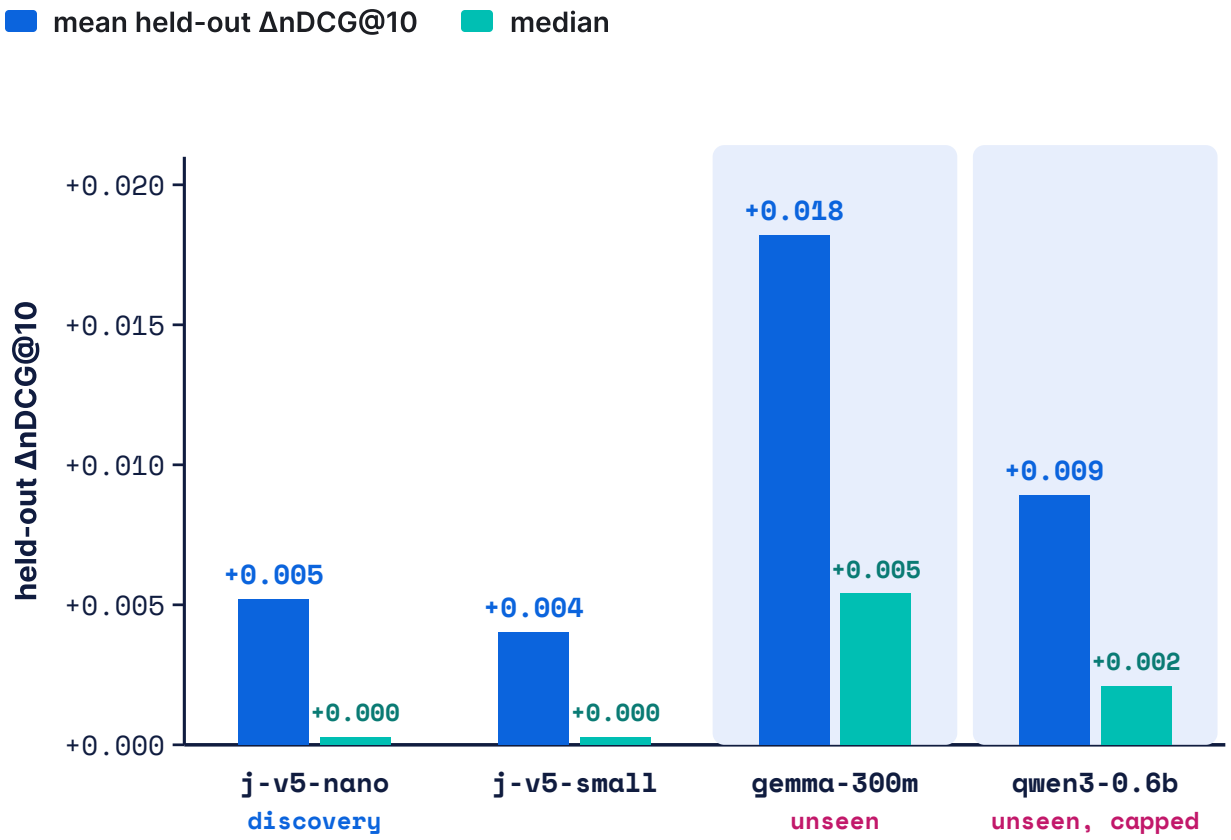
12 programs × 19 held-out tasks × 4 encoders. 485 of 912 cells improve, the worst falls to −0.98, and the pooled mean is −0.016.



52 of 76 task-encoder pairs improve under at least one program.

A transfer search finds cheap programs that generalize

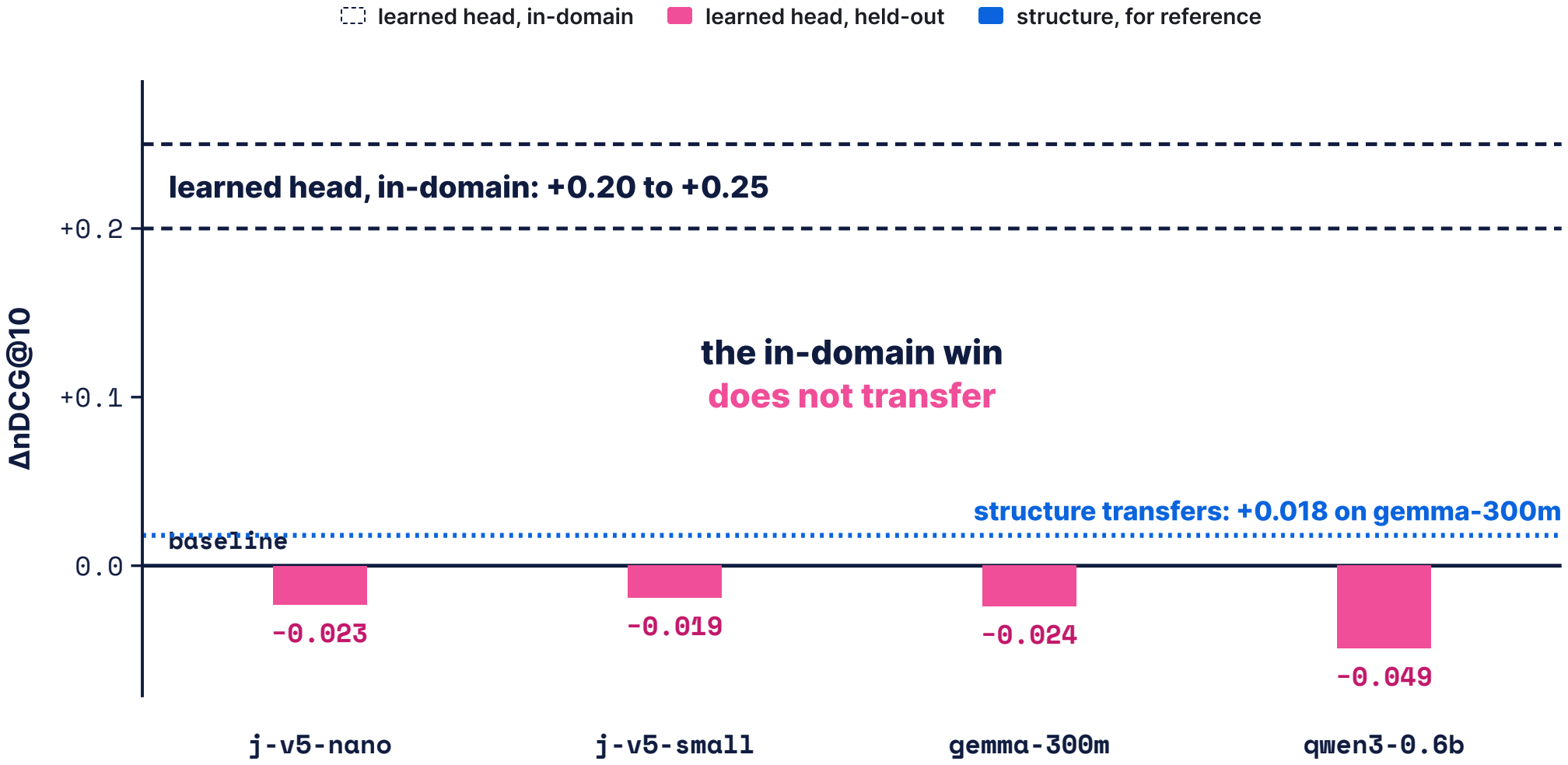
A program is kept only if a held-out validation split improves and no task drops by more than 0.05. Six pass, all at 1.5x or less.



- Mean gains are positive on all four encoders and largest on unseen families.
- Transpose-Consensus never loses a task: **83% win rate**, worst cell **0.000**.
- Applied unmodified to French and Greek: median **+0.016**, **86%** win rate.

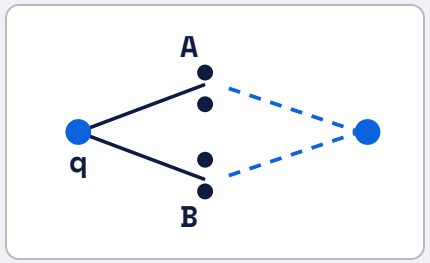
A learned head at the same data budget memorizes

Linear, low-rank and MLP heads trained on the 14 discovery tasks fall below baseline on every held-out encoder.



The search re-derives classical IR in embedding space

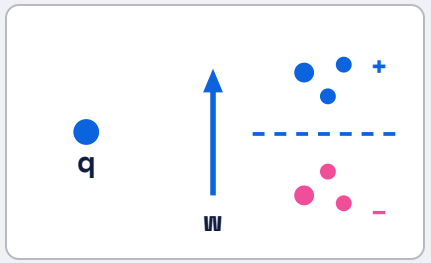
All four rely only on cosine geometry, which every encoder shares.



Reciprocal Rank Fusion

appeared without a seed

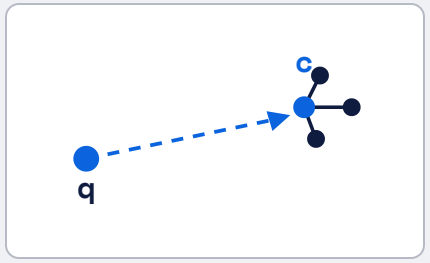
fuse two rankings into one



Fisher Linear Discriminant

appeared without a seed

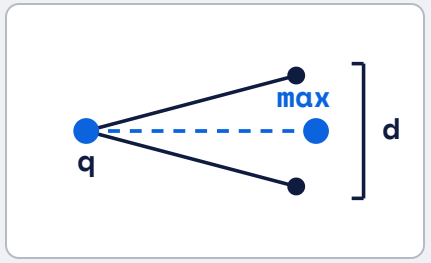
the axis that best separates relevant and non-relevant



Rocchio Pseudo-Relevance Feedback

operationalized from a seeded idea

pull the query toward the relevant centroid



Sentence-level MaxSim

operationalized from a seeded idea

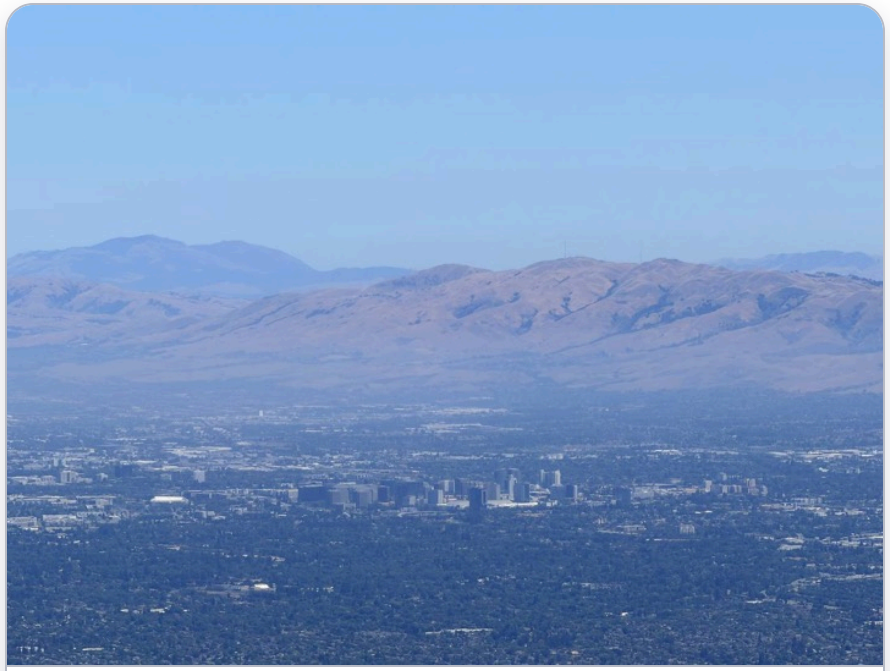
score a document by its best sentence

02

A retrieval model tags images

Open-vocabulary tags from a frozen embedding model

The tags are verbatim, with no training, dictionary or second model.



mountain · phoenix · sky · rooftop · location



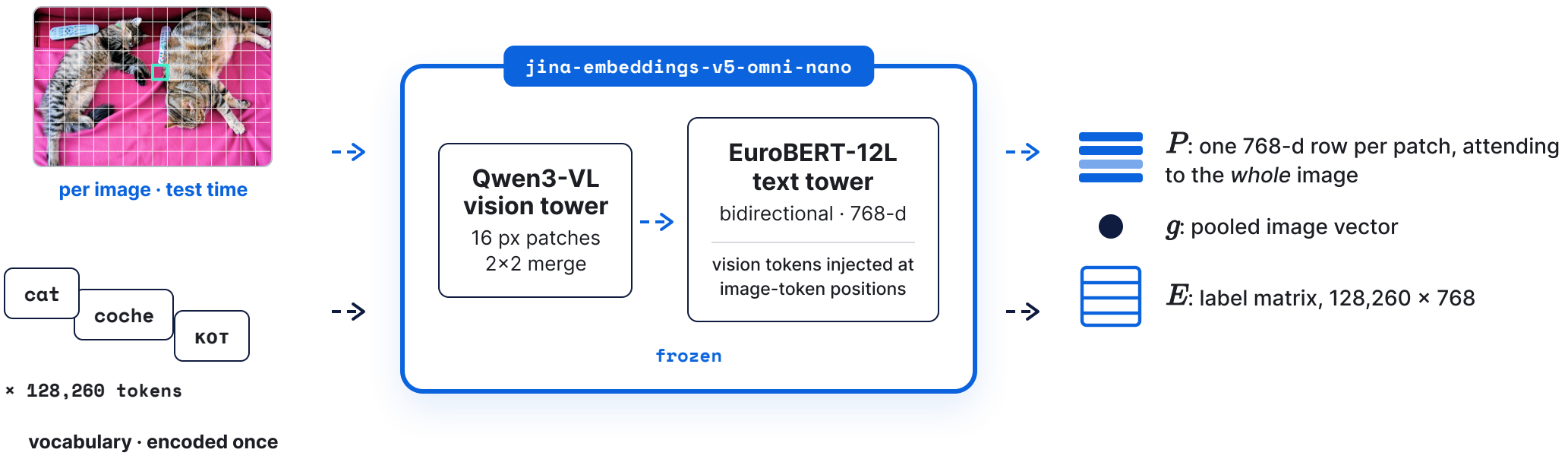


clock · thermostat · dial · dst · wifi

9 photos · top-5 tags each

Images and text share one embedding space

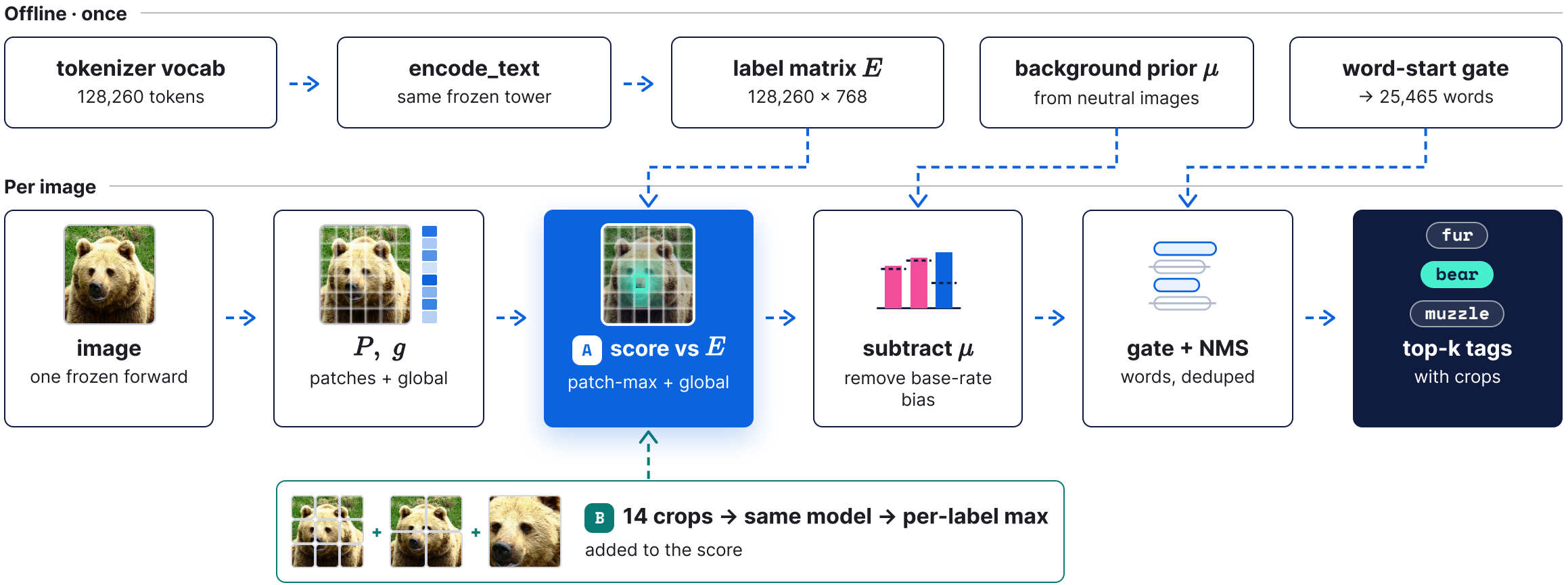
jina-embeddings-v5-omni-nano is a retrieval encoder of about 1B parameters with no tagging head.



Any image patch can be scored against any word in the vocabulary.

The whole tagger is test-time compute

The label set is the tokenizer vocabulary: 128,260 tokens, gated to 25,465 whole words.

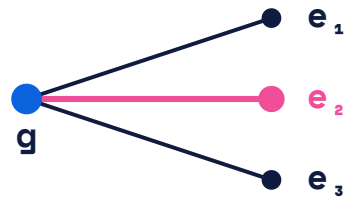


Every box is the frozen model or arithmetic. A reads more of one pass, B adds passes on crops.

Scoring every patch lifts mAP from 0.264 to 0.635

The patch vectors already exist in the forward pass. Each label keeps its best patch, as in ColBERT.

global pooled

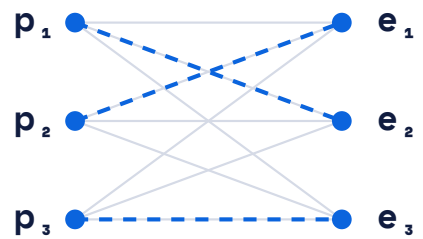


$$s_\ell = \langle g,\, e_\ell \rangle$$

one vector, dominated by the most salient object

mAP 0.264

per-label max over patches



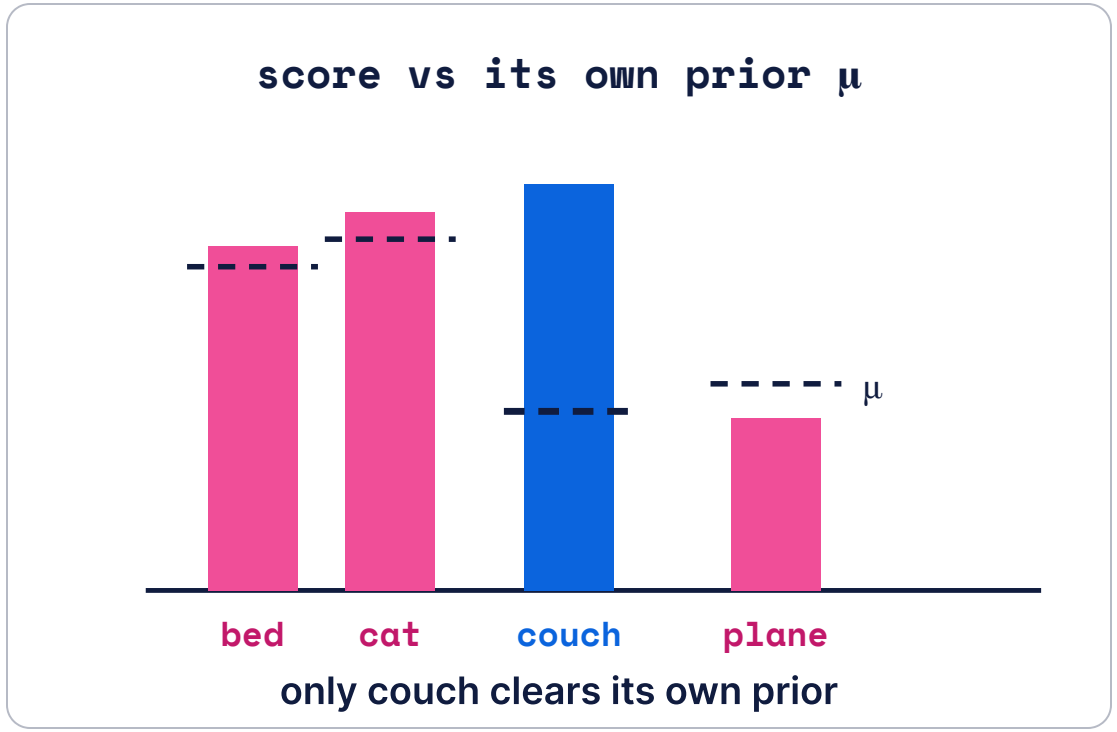
$$s_\ell = 0.7 \max_{p \in P} \langle p, e_\ell \rangle + 0.3 \, \langle g, e_\ell \rangle$$

best patch per label, plus the global vector

mAP 0.635

Each label is compared with its own prior

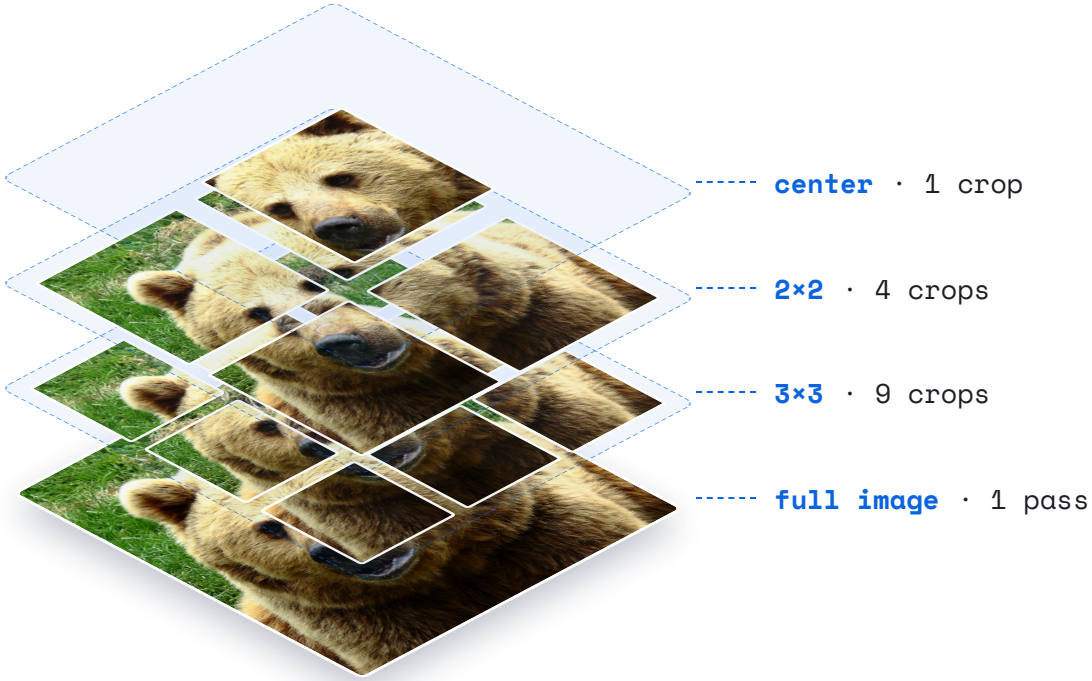
Generic words such as bed and cat score high on every image. A prior from neutral images removes that bias.



- One offline pass, with no calibration set or labels.
- **Word-start gate:** keep tokens that the tokenizer marks as word starts.
- **Embedding-NMS:** drop any word within cosine 0.6 of one already kept, so cat, kitty and кот merge.

Crops give the model new pixels

A small object fills its own crop, so weak evidence becomes strong. Each label takes its max over 14 crops.



1 pass	wolf	roar	muzzle
+ 14 crops · per-label max	fur	bear	muzzle

- **+0.075 mAP** for 14x the passes: 0.635 to 0.710.
- P@1 goes from **0.753** to **0.813**.
- One pass reads **wolf, roar, muzzle**. With crops it reads **fur, bear, muzzle**.

The whole tagger is one recursive scoring function

Recursion: the same scoring function runs again on each crop as a new image

$$S(\ell) = \underbrace{0.3\left(\langle g,\, e_\ell\rangle - \mu_\ell^g\right)}_{\text{C Calibration}} + \underbrace{0.7\left(\max_{p\in P}\langle p,\, e_\ell\rangle - \mu_\ell\right)}_{\text{A Patch evidence}} + \underbrace{1.3\left(\max_{c=1..14} s_\ell(\text{crop}_c) - \mu_\ell\right)}_{\text{B Multi-crop re-encode}}$$

C Calibration
The pooled vector minus its own background prior, at no extra cost.

A Deeper pass
Arithmetic on patch rows the forward pass already produced, at no extra cost.

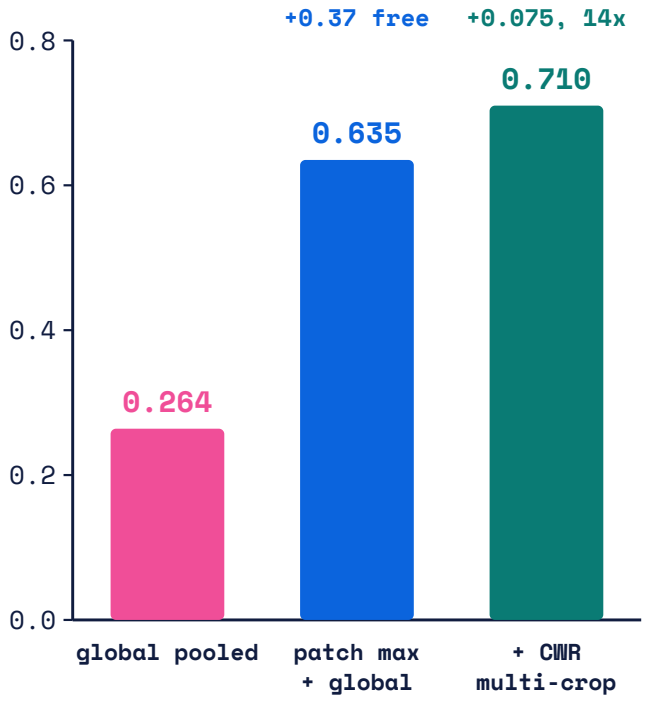
B More passes
14 new forward passes over new pixels.

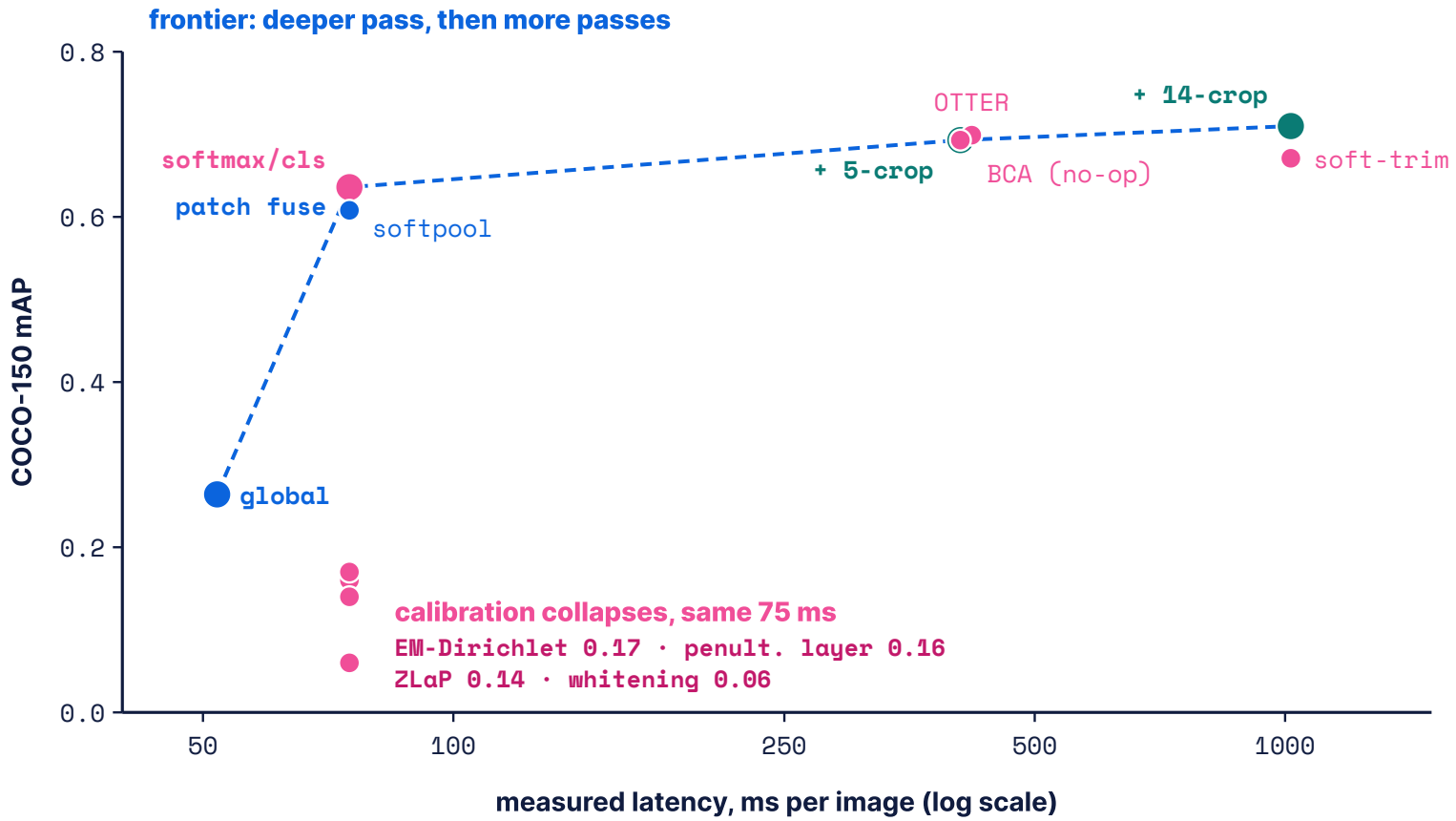
$\ell \in$ word-start gated vocabulary $\rightarrow$ embedding-space NMS $(\tau = 0.6)$ $\rightarrow$ top-k

The tagger has no classifier head, logits or learned thresholds. It uses three fixed weights, two background priors and a few max operations.

Accuracy climbs with each kind of new information

Patches give the big step. Crops give the next one at higher latency.

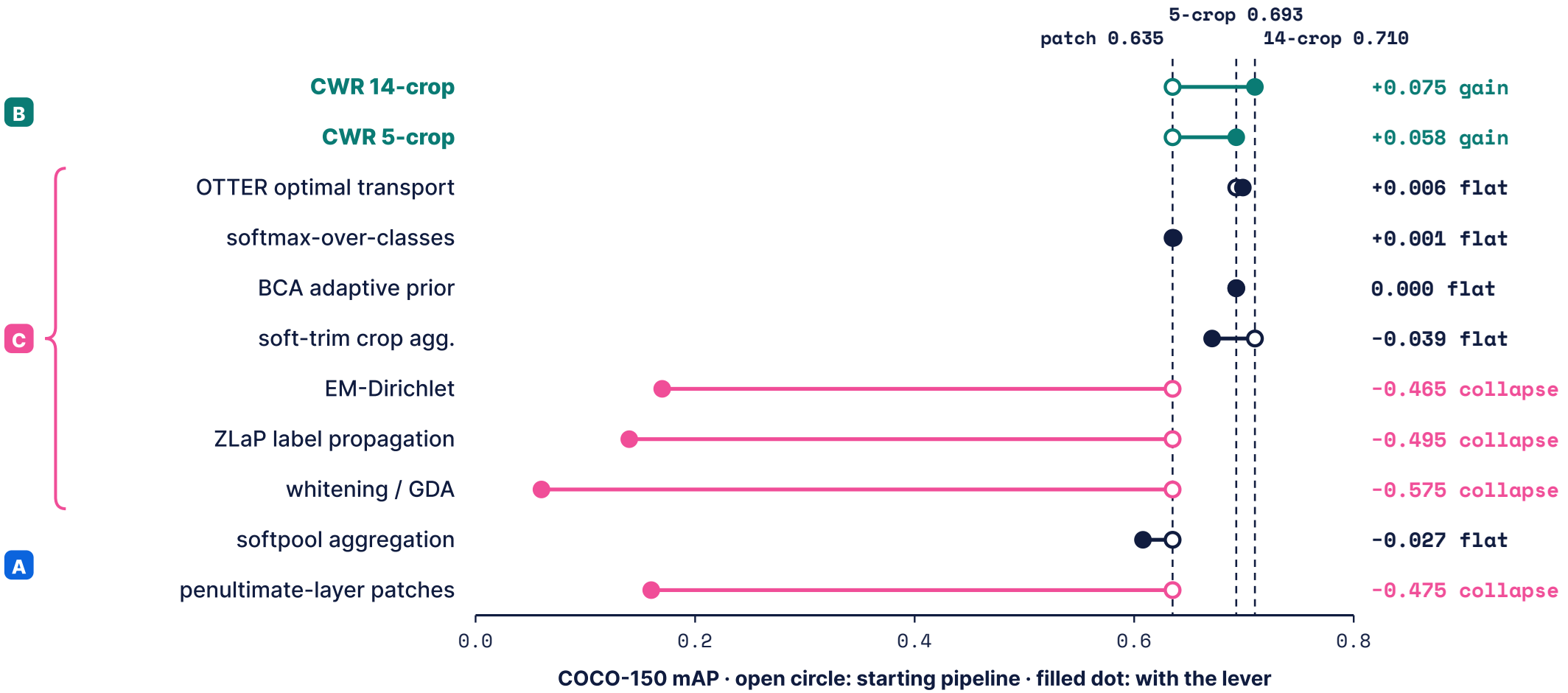




COCO-150 mAP: global → patch → + 14 crops

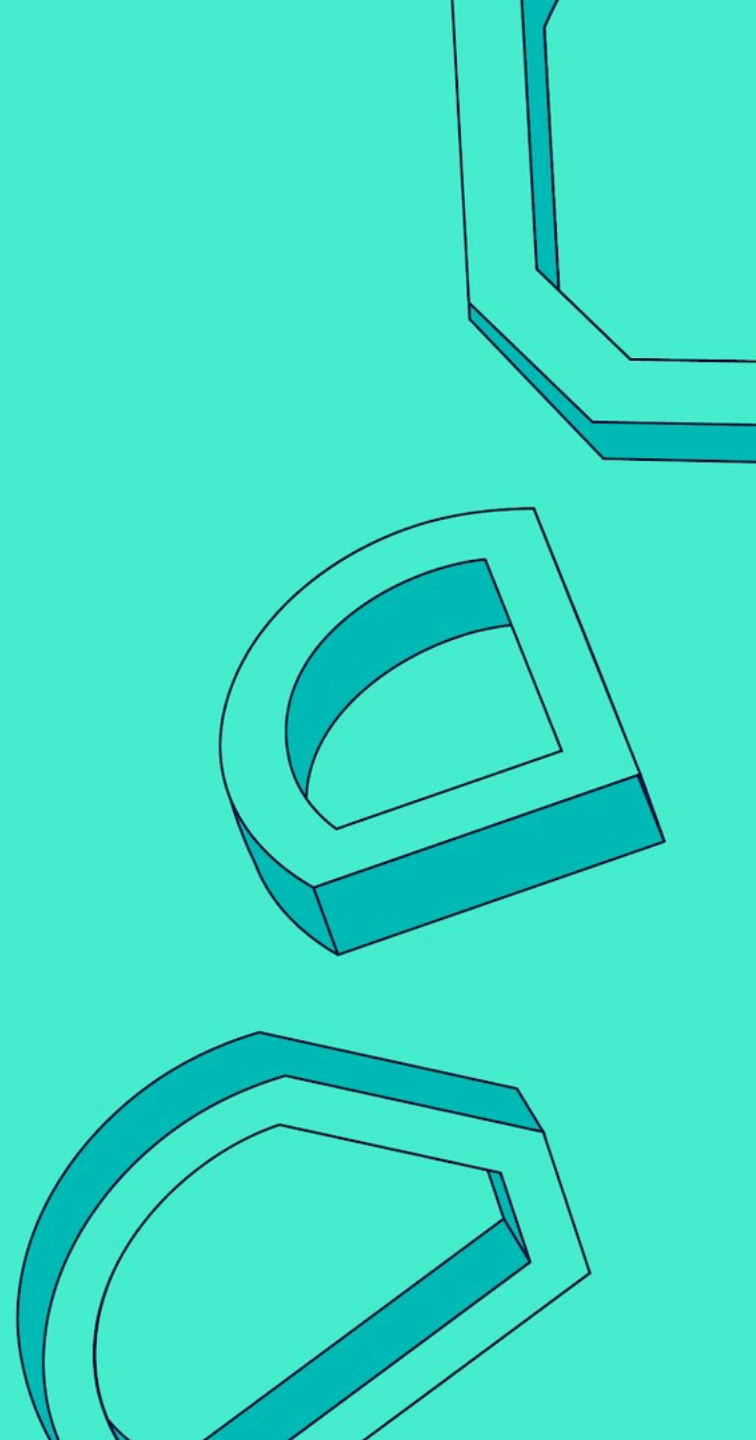
Calibration does not move accuracy

Recent training-free methods rely on calibration. On this encoder, every calibration lever is flat or collapses.



Only **new information** moves accuracy.

Patches and crops add information. Calibration rearranges what is already there.



omni-macos

Tagging runs on device in Omni

It reuses the patch rows of the embedding pass. Tags become search snippets for images, video and scanned PDFs.

~0.6 ms

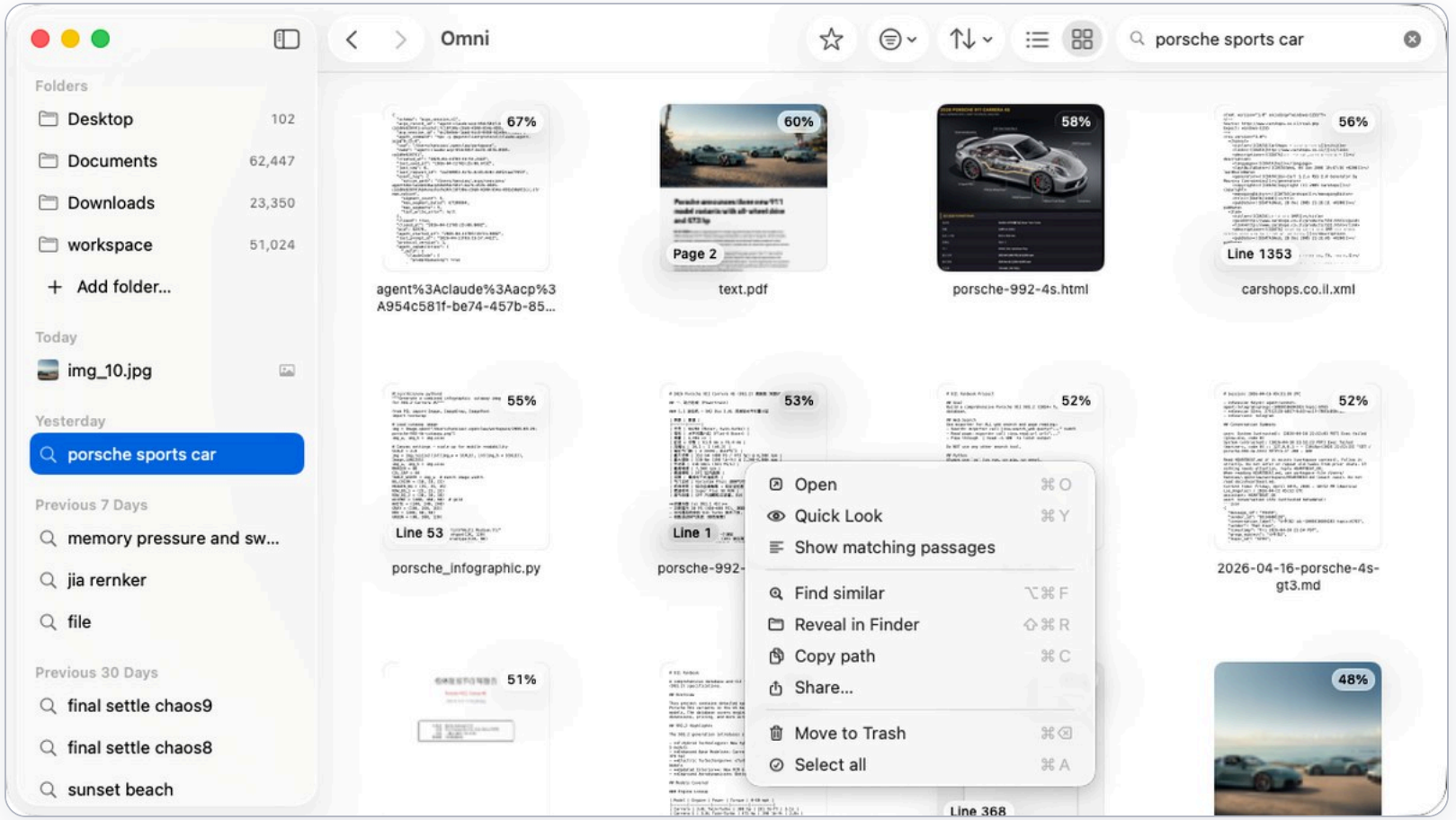
per image

~4%

of the embedding step

64

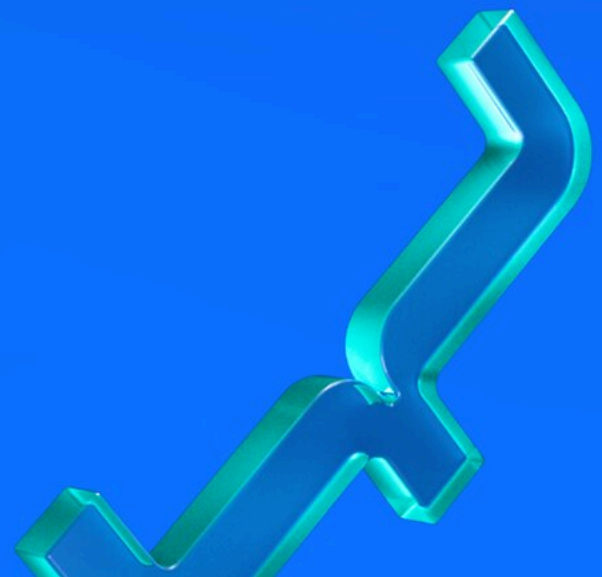
images calibrate the prior



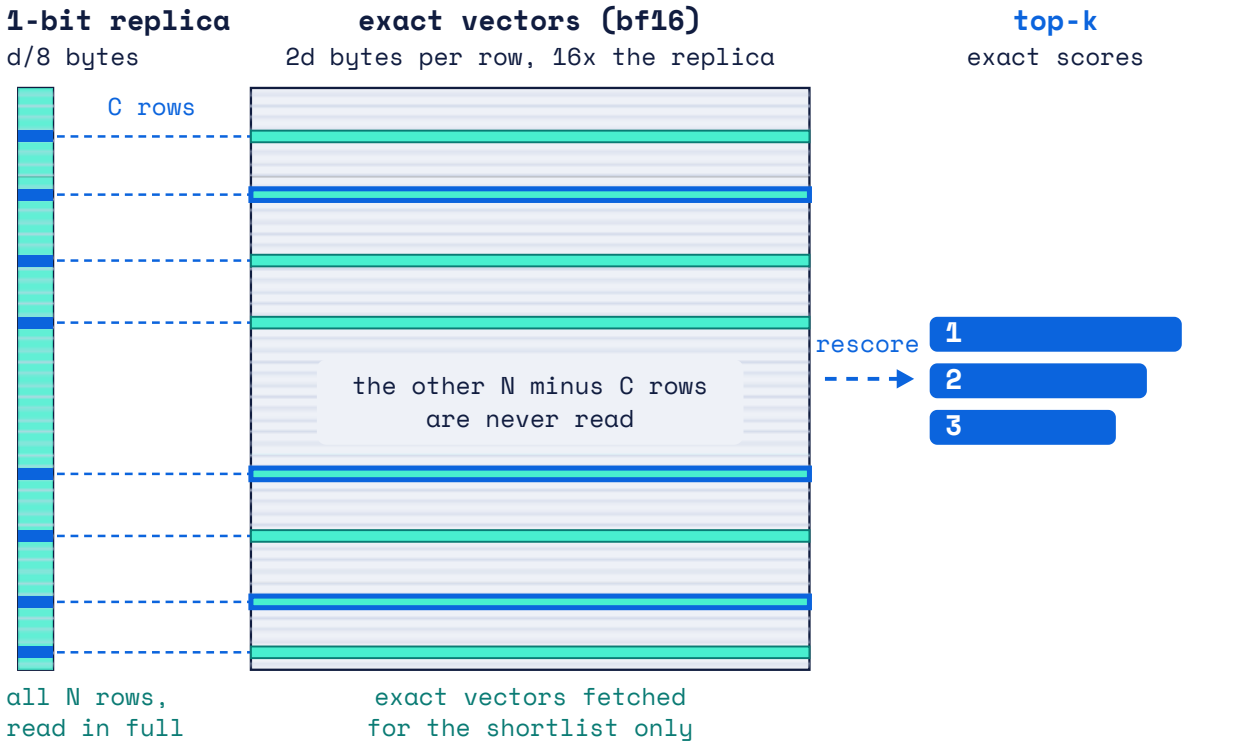


03

One call replaces an index



Omni builds no index and spends compute on rescoring



01 SCAN	02 SHORTLIST	03 RESCORE	04 TOP-K
every row, 1 bit	top C, coarse	exact, C rows	exact order

Teal marks what one query reads.

Every query scans the whole corpus. A graph index would add 256 bytes of links per row, **0.95 GB at four million chunks**, close to three times the scanned matrix, and edits keep invalidating it.

The scan reads a 1-bit replica. Exact vectors are read only for the shortlist and set the final order.

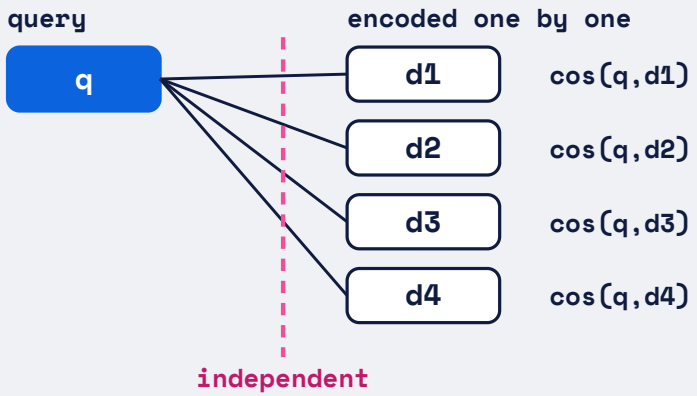
Shortlist size: `min(4096, max(1024, topK×32))`

Let the candidates see each other in one forward pass

jina-reranker-v3 (0.6B) reads the query and up to 64 documents in one context.

Bi-encoder

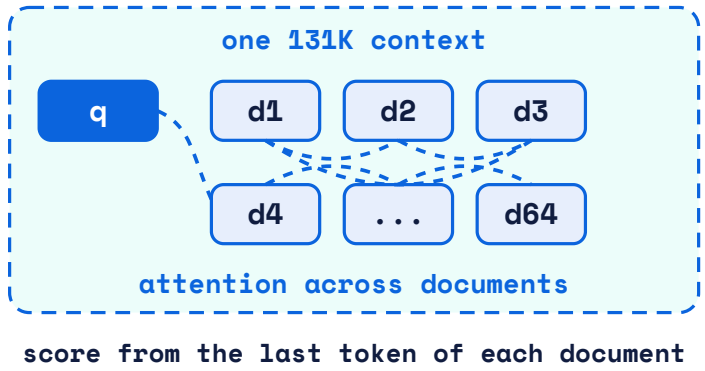
each document scored alone



Each document vector is computed without the others.

Listwise

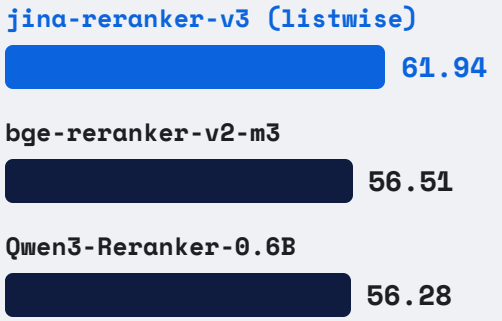
all candidates in one context



Each document is scored after attending to every other candidate.

BEIR nDCG@10

0.6B class



At the same size, the listwise model leads.

GitHub

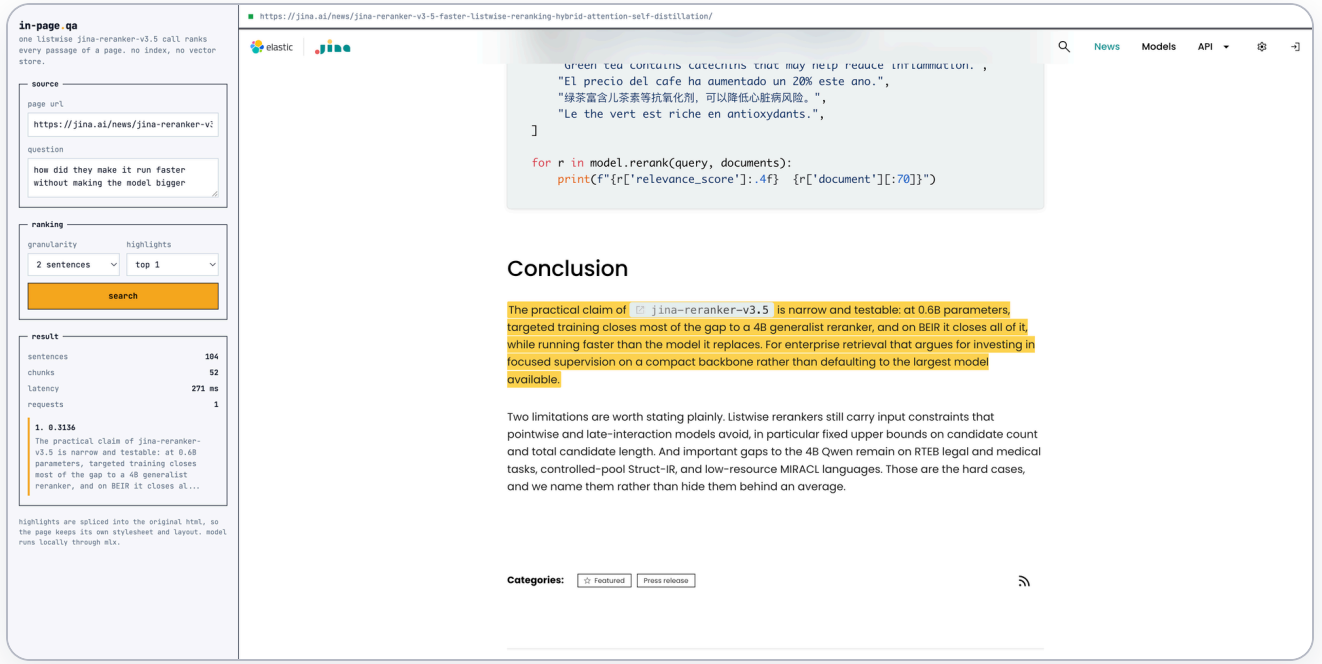
A single page needs no index

An index pays off over many queries. A page lives for one visit.

- jina-reranker-v3.5 scores a 53-chunk page in **143 ms** per call.
- Answers are highlighted in place on the original page.

ms per call · 53-chunk page

jina-reranker-v3.5	143
jina-reranker-v2-base-multilingual	4,194



04

An agent builds the pipeline

Test-time compute is the pattern for long-horizon agents

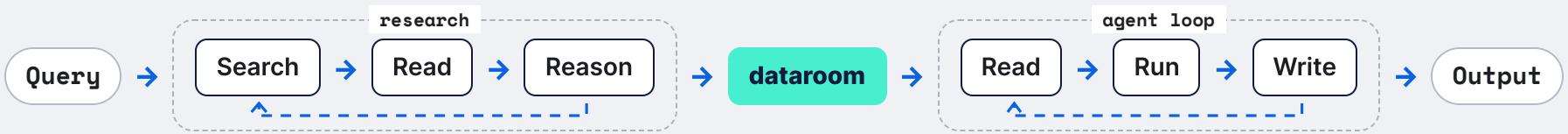
Cheap local tokens build the corpus. Frontier tokens go to the work that needs them.

2025 · Deep research



one loop · web IO throughout

2026 · Long-horizon tasks

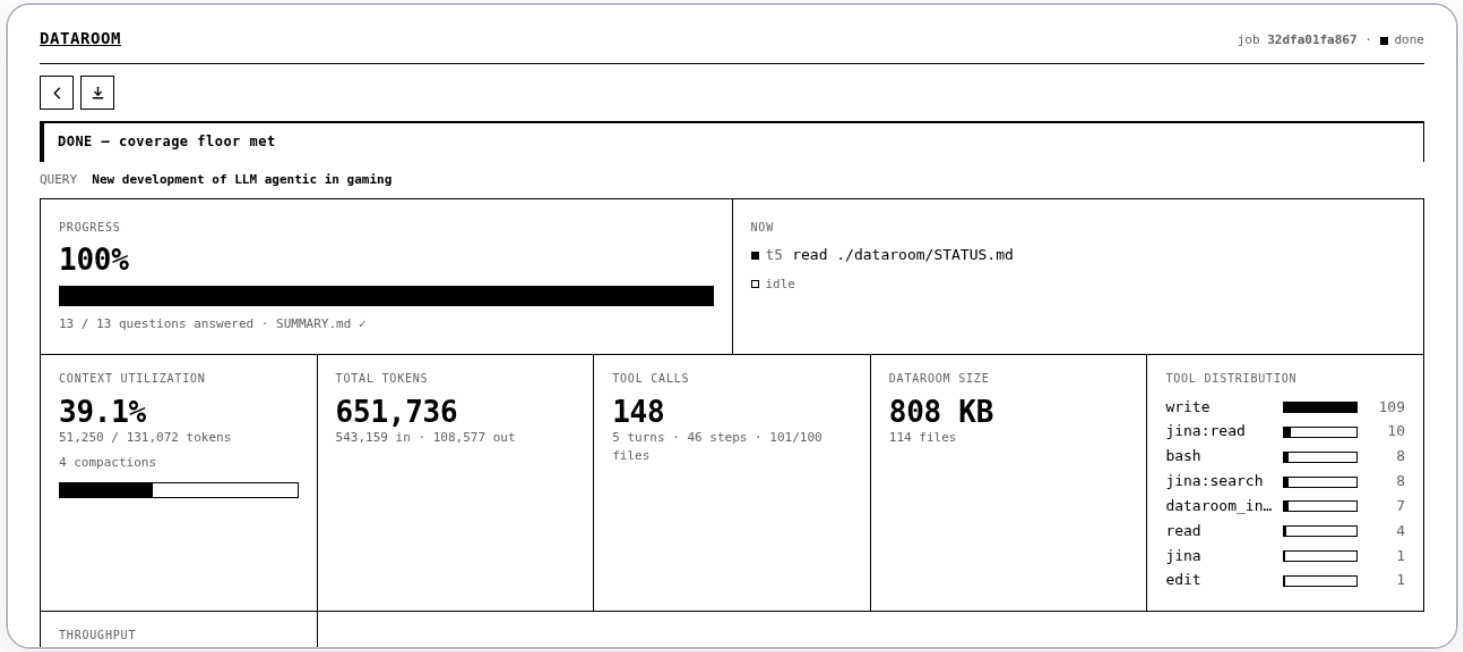


research with web IO → corpus → offline agent loop, no web IO

dataroom: a token budget builds a corpus

A local model searches, reads and writes until the knowledge sits in one cited .zip.

GitHub

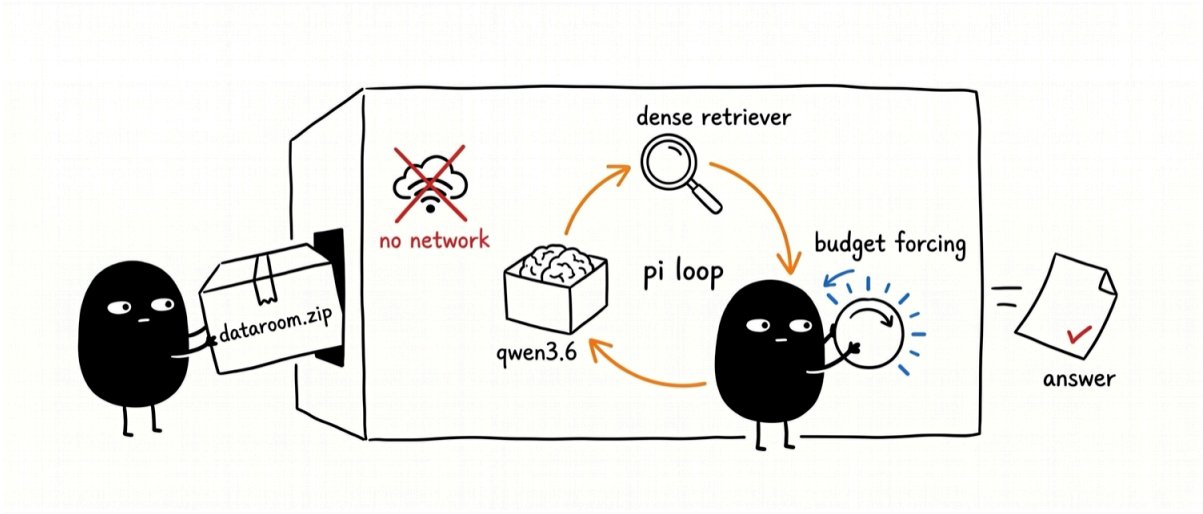


- It stops at a measured coverage floor.
- The job page tracks coverage, tokens and tool calls.
- The .zip goes to searchbox or to a frontier model.

GitHub

searchbox: an agent builds its pipeline for each query

The agent gets one dataroom, local tools and no web access.

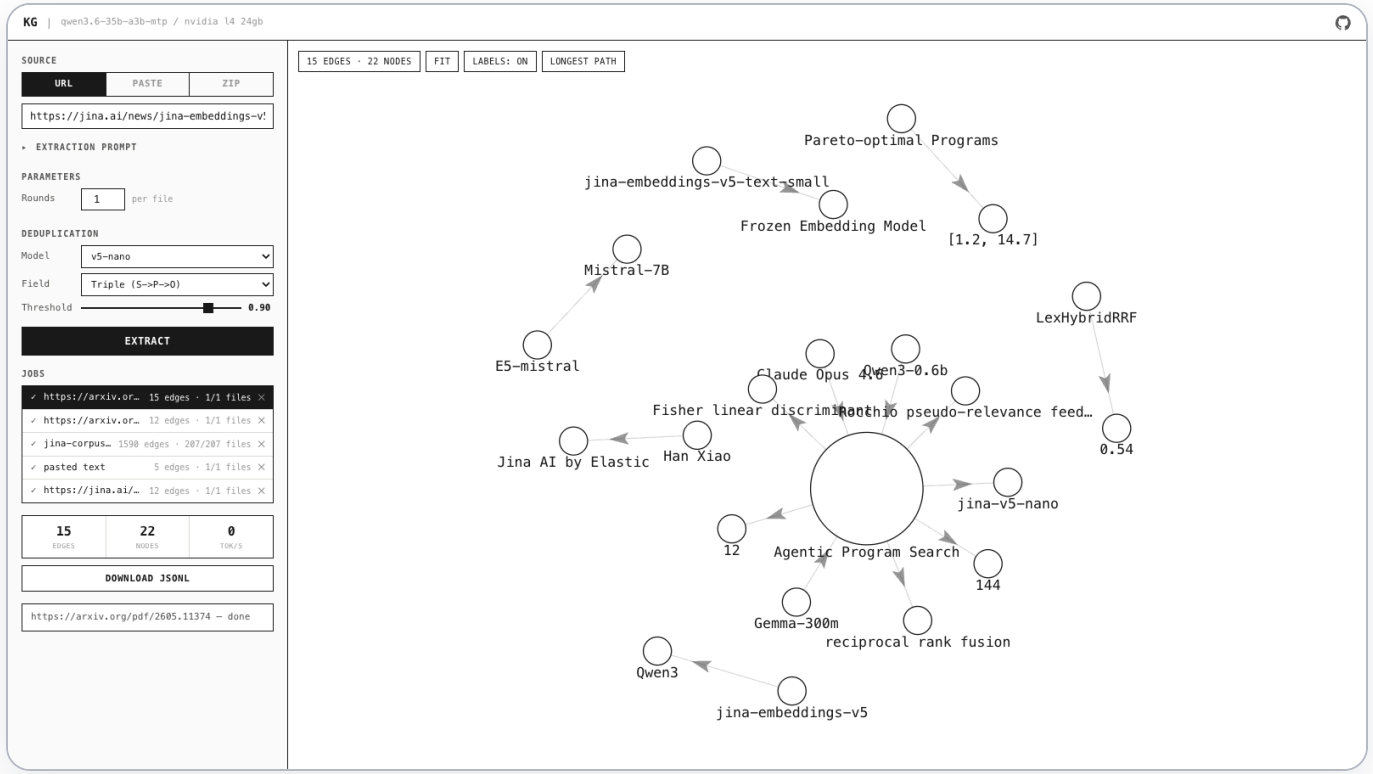


- **Tools:** grep, embed, rerank, similarity, cluster, select_diverse.
- → Which tool does the agent call first?
- → When does grep match a dense retriever?
- → Does a larger token budget help on the hard questions?

GitHub

knowledge-graph: hard questions as a private verifier

If one grep finds the answer, every method scores the same. Agentic search needs questions that force several hops.



- Turn the corpus into subject-predicate-object edges.
- Walk paths of length 2 and 3, hide the bridge entities and ask for the endpoint.
- Keep a question only if a closed-book probe fails and a retrieval agent succeeds.

Lessons from the four demos

Training-free

Programs, tags, listwise ranking and agent pipelines all run on frozen weights.

Compute buys quality

Programs gain +0.24 in search, a retrieval model tags images, and one call replaces an index.

New information drives gains

Patches add +0.37 mAP at no cost, crops add +0.075 at 14x, and calibration adds about zero.

Check out of domain

+0.24 in search became −0.016 on held-out tasks. Cheap programs and private verifiers pass this check.

Test-time compute in Elasticsearch today

`retriever · rrf`

Fuse rankings

Fuse lexical and dense results with the `rrf` retriever, which the program search rediscovered.

`retriever · text_similarity_reranker`

Rerank the top-k

Rerank the top hits with a listwise model through the `text_similarity_reranker` retriever.

`field · semantic_text`

Score chunks

With `semantic_text` chunking, rank each document by its best chunk.

`agents · budget`

Budget the agent

Give agents a token budget, and test them on private multi-hop questions before raising it.

Search has always been test-time compute. Spend it where **new information** enters.



Thank you

Han Xiao, VP of AI, Elastic

